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Quadrilaterals

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Quadrilaterals

A **polygon** is a closed, plane (flat) figure made up of line segments. In this case, *closed* means that the sides meet.



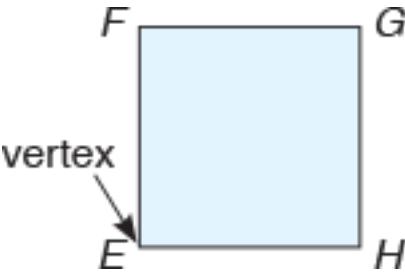
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not closed

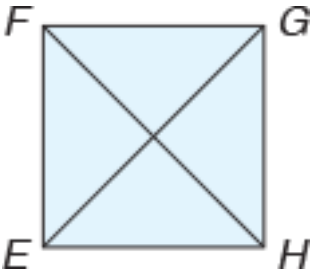
A **quadrilateral** is a polygon with four sides. Squares and rectangles are the two most common quadrilaterals. If you add the angles inside any quadrilateral, the sum is always 360° .

Look at the quadrilateral *EFGH* shown here. Each time two sides meet, an angle is formed. The point where the sides meet is called a vertex. The four labeled points are the **vertices** of this figure.



Sides *EF* and *GH* are **opposite** (across from) each other. Sides *FG* and *EH* are also opposites. Sides *EF* and *EH* are **adjacent**. This means they share an endpoint.

A straight line connecting opposite vertices (corners) is called a **diagonal**. Line segment *EG* is a diagonal; so is *FH*.



Below are examples and properties of five common quadrilaterals.

Name	Properties	Examples
square	Opposite sides are parallel. All sides are equal. All angles are right angles (90°). Diagonals are equal and perpendicular.	
rectangle	Opposite sides are equal and parallel. All angles are right angles (90°). Diagonals are equal but not necessarily perpendicular.	
rhombus	<u>Opposite sides are equal and parallel</u> . All sides are equal. Diagonals are perpendicular but not necessarily equal.	
parallelogram	Opposite sides are equal and parallel.	
trapezoid	Only one pair of sides is parallel.	

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Lesson 1: Basic Figures

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